PROGRAMME DE RECHERCHE

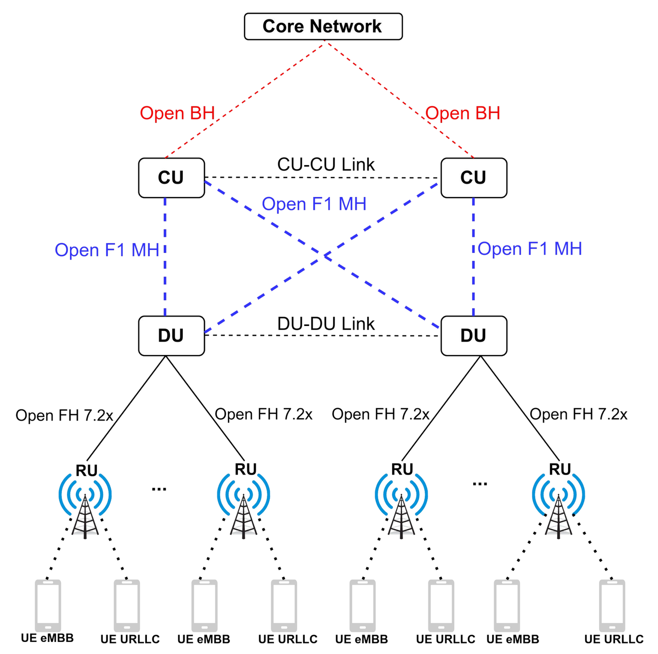
RÉSEAUX DU FUTUR

André-Cédric BESSALA, Guilherme IECKER RICARDO and Gentian JAKLLARI (PhD 1.7)

PEPR NF-NAI

E-MATRO : Adaptive and Energy-Efficient Framework for Slicing in O-RAN Architecture

Context



O-RAN network architecture.

O-RAN is an industry initiative to create open, intelligent, virtualized and fully interoperable mobile networks. As a benefit, it brings MNOs :

- Modular hardware/software components,
- Functional splits (RU, DU , CU),
- Programmable RAN intelligent controllers (RIC).

Problem Definition

Dynamic traffic and strict QoS requirements challenge traditional centralized control in O-RAN networks, resulting in suboptimal resource allocation and high energy consumption. This work addresses the issue of energy-efficient, slice-compliant VNF placement, formulating it as a flow-based optimization model as described bellow.

$$\min_{x,y} E(x,y) = \sum_{i\in V} \epsilon_i \cdot \left[W_i^V(x) + W_i^F(y)\right]$$

subject to:

$x_{r,l}^{VNF} \in \{0,1\}$, $\forall r \in V^{RU}$, $\forall i \in V^{Cloud}$, VNF $\in \{DU, CU\}$

$y_{r,s,l} \in \{0, 1\}$, $\forall r \in V^{RU}$, $\forall l \in L$, $\forall s \in S$

$W_i^V(x) + W_i^F(y) \le \mu_i^{Comp}$, $\forall i \in V^{Cloud}$ //Virtualization and forwarding computing cost

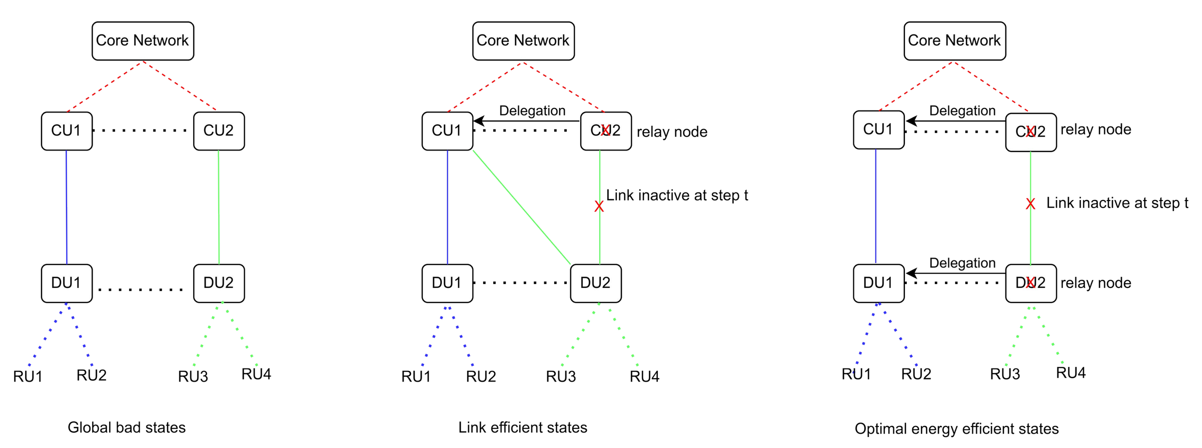
$L_l(y) \le \mu_l^{Link}$, $\forall l \in L$ // Link capacity

$D_{r,s}(y) \le \delta_r$, $\forall r \in V^{RU}$, $\forall s \in S$ // End-to-end delay

→ **This problem is NP-Hard !**

Example

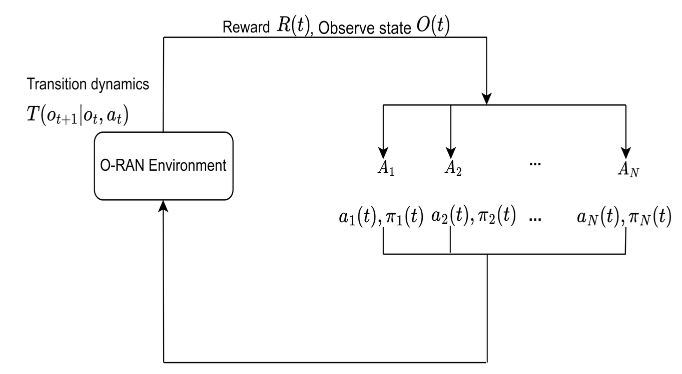
Consider the O-RAN network as shown above. To improve the efficiency of this system, a MARL approach can converge to three solutions.



Example solutions of the proposed ORAN architecture.

- Global bad states : In this configuration, the MARL utilizes the full spectrum of resources in the network.
- Link efficient states : To guarantee QoS for users' slices, we incur costs associated with avoiding unfeasible states.
- Optimal energy efficient states : If there is sufficient capacity on a specific path to support all traffic, the optimal decision is to migrate all processing from other paths to this one.

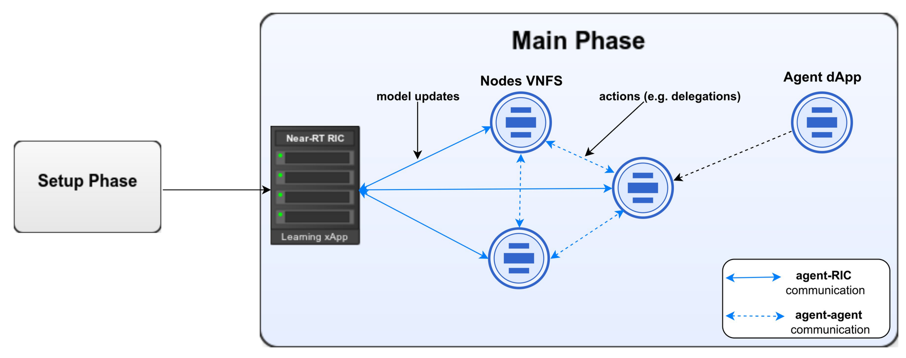
Multi-Agent RL



Multi-Agent RL framework.

- **States :** The state $o_i(t) \in O$ represents the observed status of each agent at time t.
- **Actions :** Each agent $i \in V$ can learn two distinct types of actions at time t : Function migration and traffic steering.
- **Transition dynamics :** The transition dynamics are determined by the collective interactions of all agents.
- **Reward :** Agents receive rewards based on actions that influence their transition to the next state.The reward accounts for total energy consumption, penalized by QoS constraints.
- **Policy :** Each agent selects the action based on the optimal policy $\pi_i^*(o_i(t))$.

E-MATRO

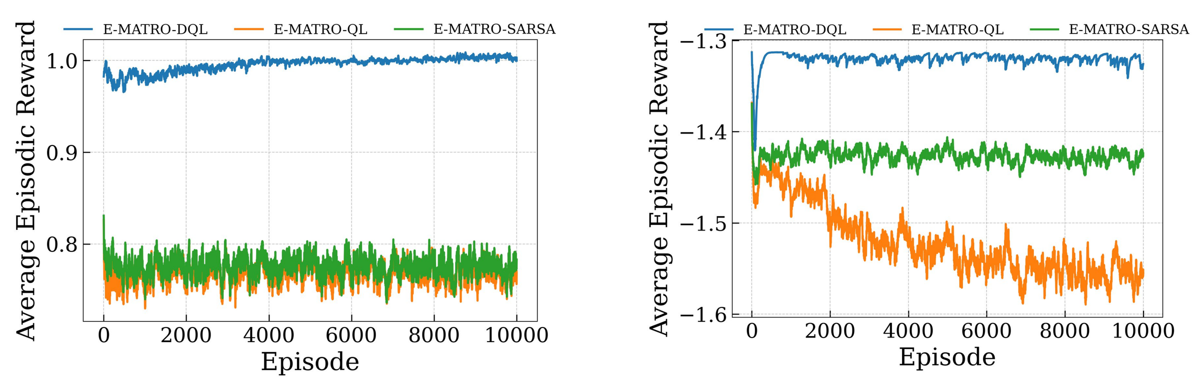


E-MATRO framework.

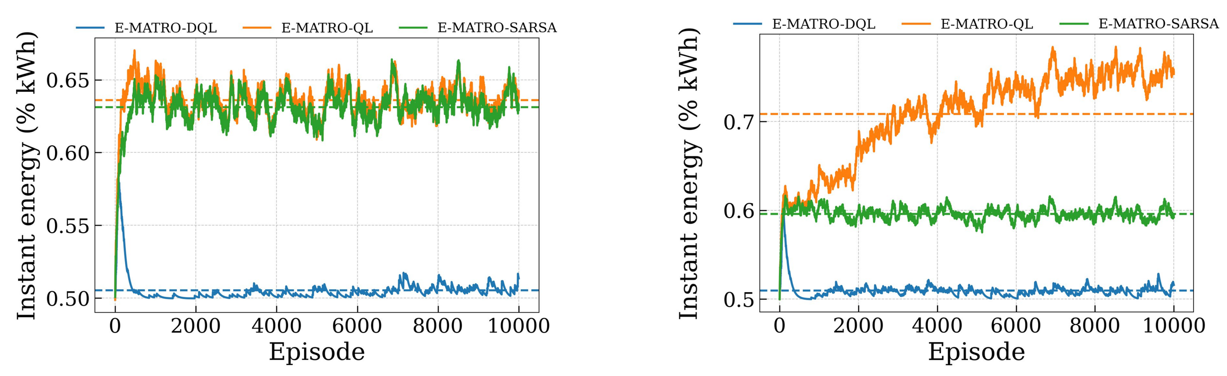
E-MATRO refer to Eco Migration And Traffic Re-routing in O-RAN, a hybrid MARL framework featuring a Near-RT RIC as the central xApp, with each VNF node in the O-Cloud network acting as a dApp. E-MATRO consists of two phases :

- **Setup Phase :** Managed by xApp, responsible for the initial system configuration and environment preparation.
- **Main Phase:** Features active dApp participation, with each dApp operating directly on its respective O-Cloud VIM as an autonomous RL agent.

Results



Reward across two different scales of topologies under light-traffic load.



Energy analysis across two different scales of topologies under light-traffic load.

- **Result1 :** E-MATRO can converge to an optimal policy across networks.
- **Result2** : E-MATRO effectively minimizes energy consumption and scales efficiently.

This work has been accepted and will be presented to IFIP networking 2025 conference.

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